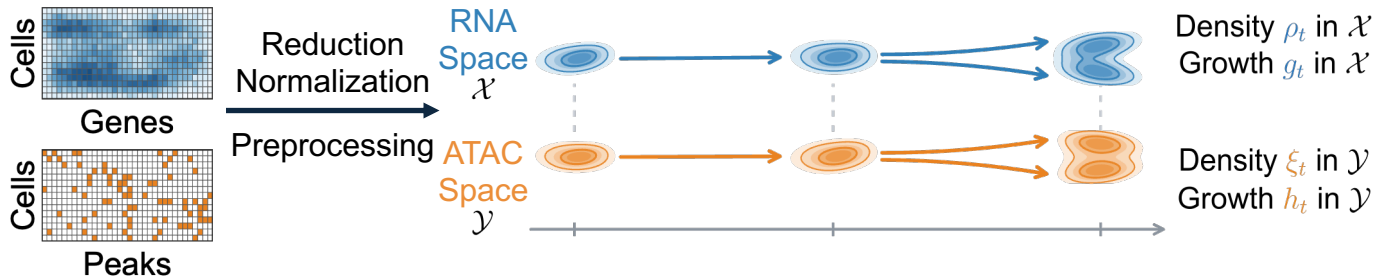
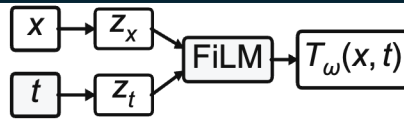
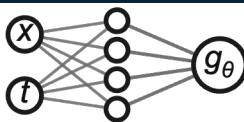
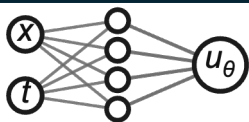


Input

Coupled Cell-group dynamics in two modalities

Output

RNA & ATAC
Embedding

Neural ODE

$$\partial_t \rho + \alpha \nabla_x \cdot (\rho u_\theta) = \beta \rho g_\theta$$

Trajectory in RNA

$$(T_\omega)_\#(\rho_t) = \xi_t$$

$$\partial_t \xi + \alpha \nabla_y \cdot (\xi v) = \beta \xi h$$

Trajectory in ATAC

- Observed cell
- Correct Prediction
- Observed distribution
- Failure Prediction

w/o L_{Ref} w/o L_{Manifold} w/o L_{Mass} COATI
Output
 (ρ_t, g_t)
 (ξ_t, h_t)

Optimize

$$\inf \int_0^1 \left[c_x \int_{\mathcal{X}} (\alpha u_t^2 + \beta g_t^2) d\rho + c_y \int_{\mathcal{Y}} (\alpha v_t^2 + \beta h_t^2) d\xi \right] dt$$

$+ L_{\text{Ref}} + L_{\text{Manifold}} + L_{\text{Mass}}$